CROSS-ASSET QUANT RESEARCH

Emerging Market Equities vs. Platinum

EEM vs. PPLT

Trade: LONG PPLT / SHORT EEM | **Universe:** Equities ↔ Commodity |
Horizon: 1-6 months | **Conviction:** Medium

EEM's tech-heavy portfolio has outpaced platinum's broader industrial-growth exposure, creating a relative-value gap that could narrow as hydrogen demand and platinum-linked emerging-market activity strengthen, while AI becomes a less dominant driver of EEM's performance.

SNAPSHOT

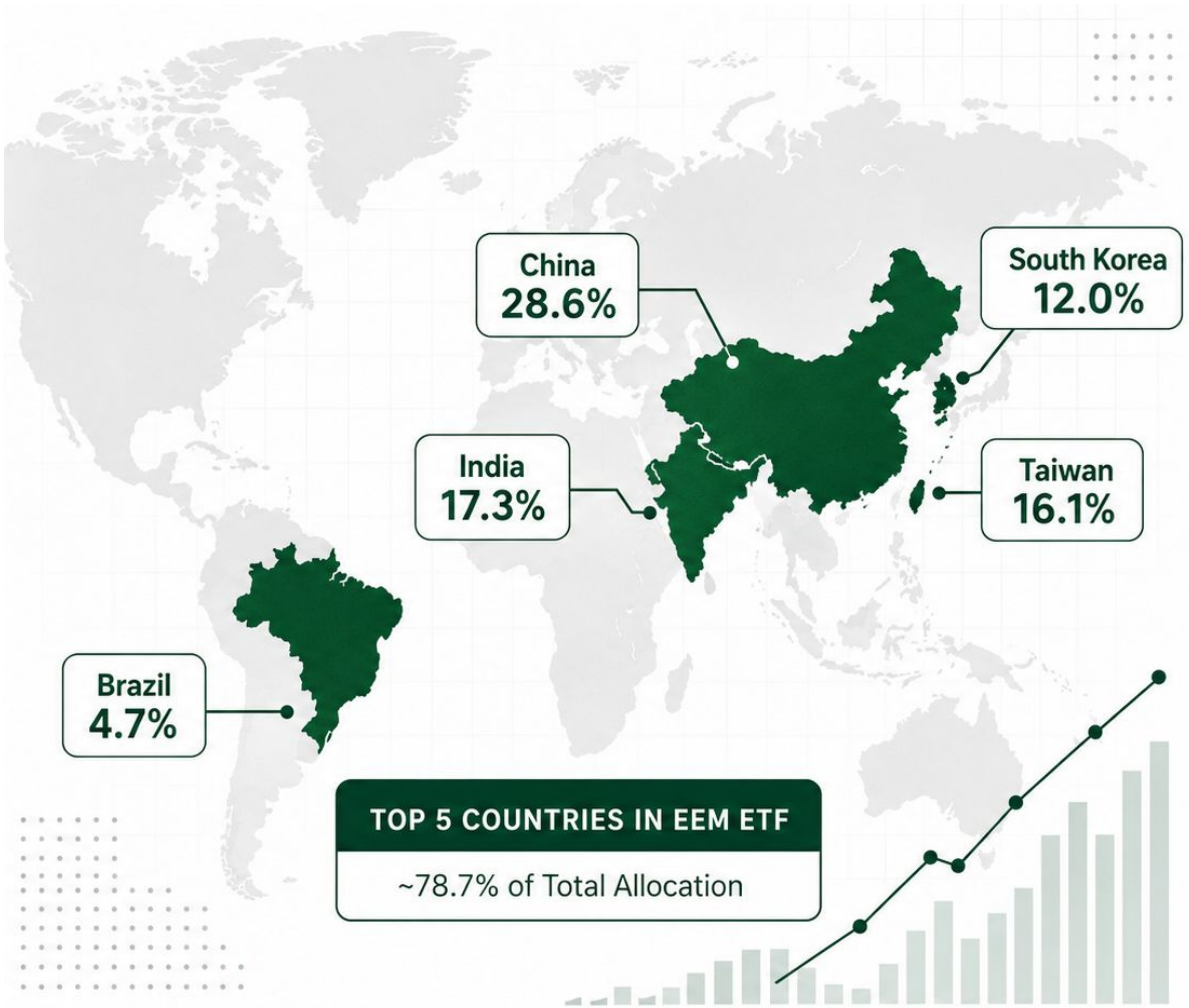
Current Gap	**+2.09σ**
Correlation (1 year)	**+0.45**
Lookback	**3-year rolling model / 16.7 years of price history**
Times This Wide	**497/3450 (14.4%)**
Long Leg	**PPLT**
Short Leg	**EEM**
Target	**0.85σ**
Stop	**2.75σ**
R/R	**1.72**

Yahoo Finance, *EEM and PPLT Historical Prices*; author's calculations.

Emerging Markets ETF

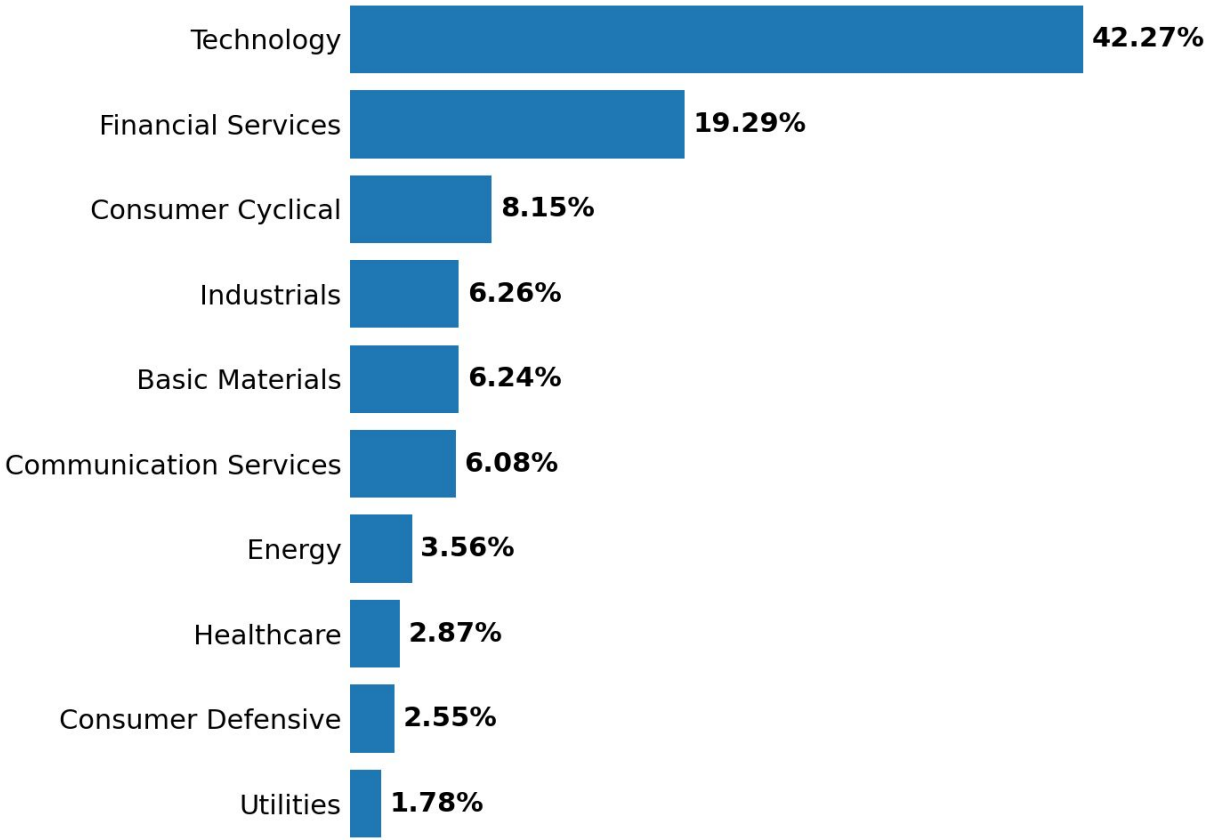
Ticker: EEM

An exchange traded fund that owns ~1186 of large and mid-sized companies from emerging markets

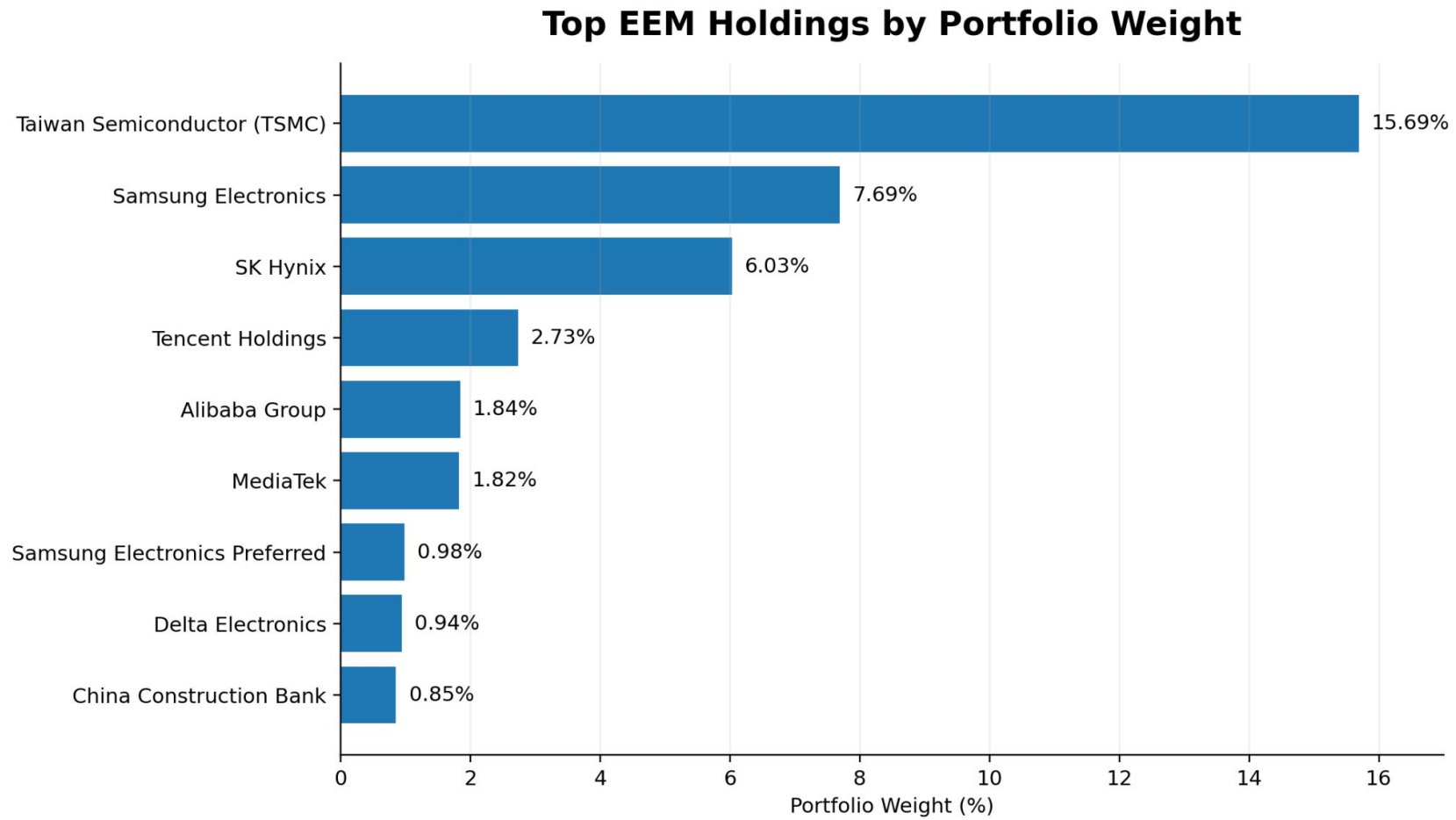


BlackRock iShares, *EEM Fund Overview*; Above the Green Line, *EEM ETF*; Aberdeen Investments, *PPLT Fund Overview*.

EEM Sector Exposure



BlackRock iShares, *EEM Sector Exposure*; Yahoo Finance, *EEM Holdings and Sector Weights*.



Hon Hai Precision Industry omitted because no portfolio weight was provided.

Yahoo Finance, *EEM Holdings*; BlackRock iShares, *EEM Portfolio Holdings*.

It is clear that one of the strongest influences on the index is AI infrastructure, as many of its top holdings are involved in advanced chips, memory, semiconductors, data centers, power systems, servers, or cloud computing–all of which support the growth and production of AI technologies.

BlackRock iShares, *EEM Portfolio Holdings*; author's analysis of AI exposure.

Platinum demand drivers



Autocatalysts

Fuel cell electric vehicles

Spark Plugs



Glass manufacturing

Carbon-free green hydrogen

Medical Devices

Hard drives, semiconductors, sensors



Rings

Bracelets

Necklaces



Physical Platinum ETFs

Platinum coins/bars

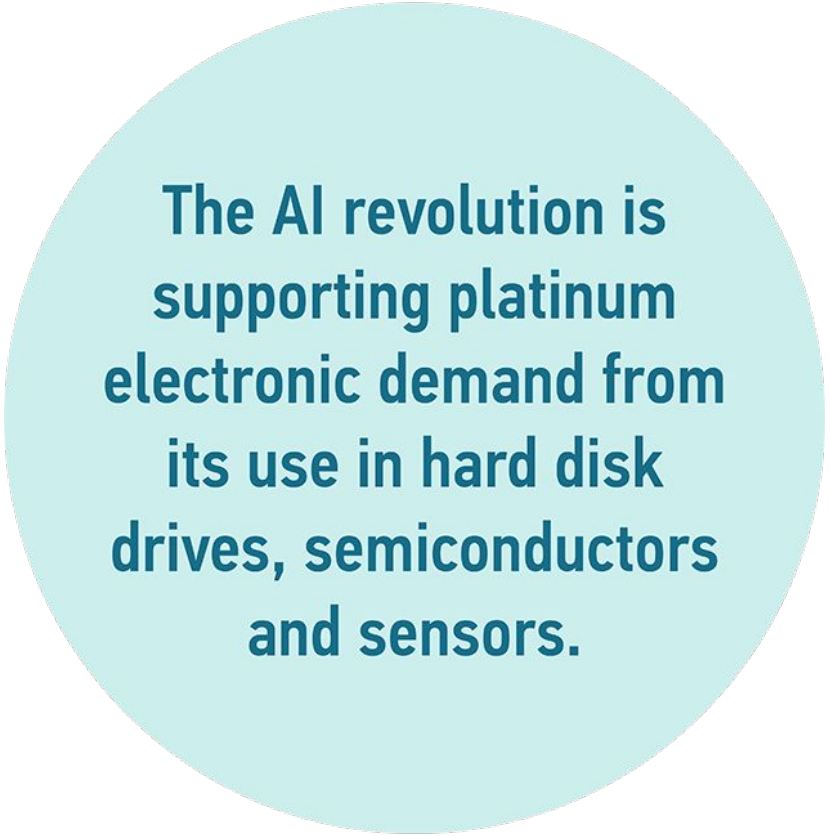
WPIC, *Platinum Demand Drivers.*

Economic Mechanism

EEM and platinum are connected because both are affected by global economic growth, technology, and AI demand. Industrial, automotive, and hydrogen demand support platinum while also benefiting parts of emerging-market economies, and South Africa creates a direct supply-side connection because platinum exports affect companies and economic activity represented in EEM.

WPIC, *Platinum Demand Drivers*; BlackRock iShares, *EEM Fund Overview*; author's analysis.

Transmission Channel 1: The AI Revolution



Electrical demand for platinum is only currently around 1% of total platinum demanded

Platinum is used across AI hardware and electronics manufacturing

About 40% of EEM is concentrated in technology and semiconductors, with TSMC alone making up roughly 15% of the fund

WPIC, *Electrical Demand* (2025), *Platinum on Film* (2024), *Platinum Platters* (2024); Yahoo Finance, *EEM Holdings*.

Transmission Channel 2: Platinum unlocks the hydrogen economy

Proton exchange membranes (PEM) are constructed using platinum catalysts

PEM electrolysers turn renewable energy into green hydrogen

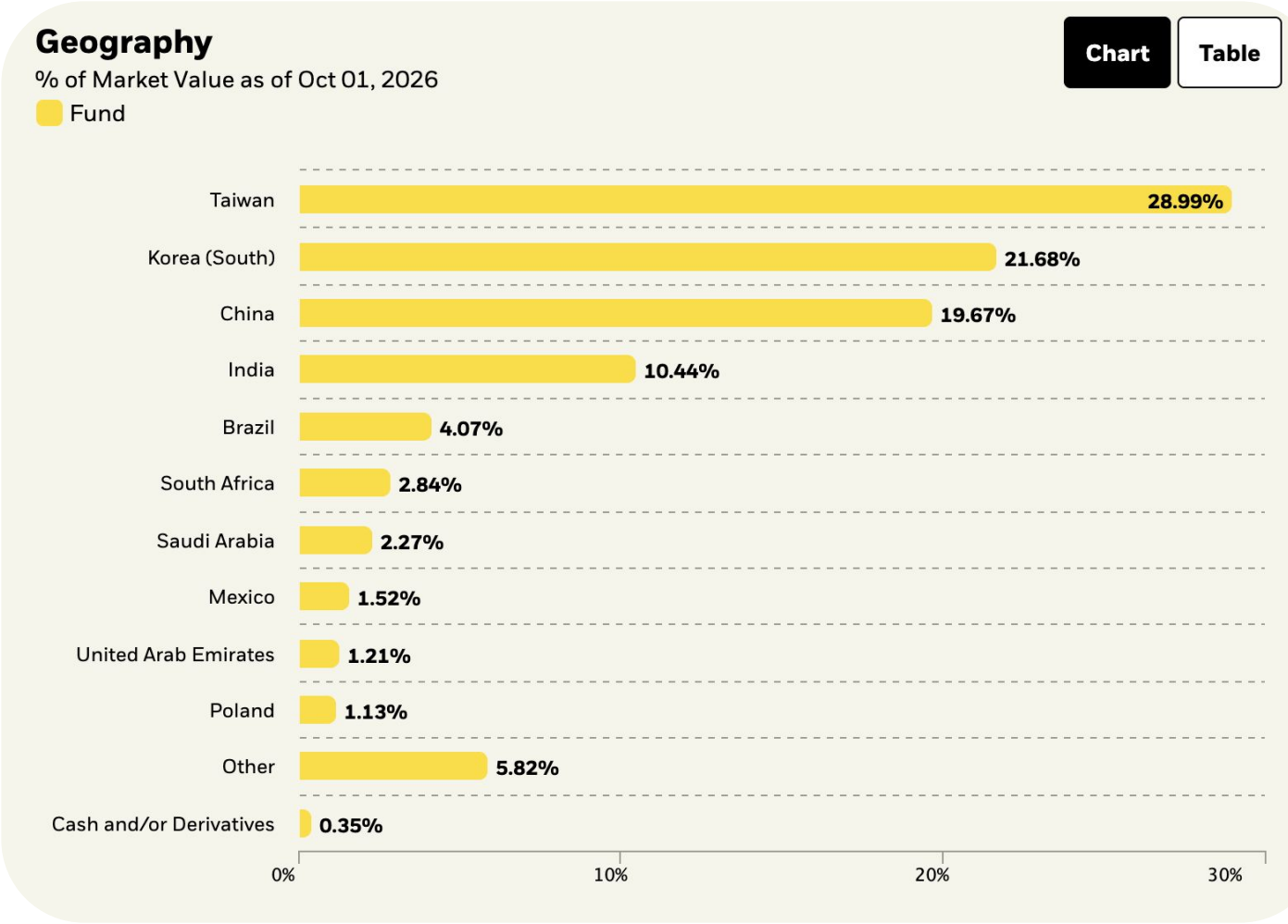
Green hydrogen is used across automotive and industrial industries, as well as to generate power for uses like data centers

PEMs are also used in creating the fuel cell for electric vehicles (FCEVs)

Hydrogen Demand for platinum is expected to grow by 18x through 2030

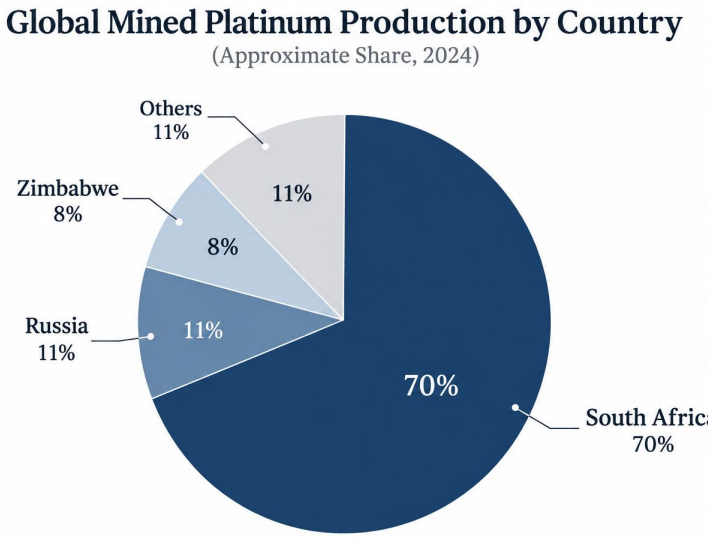


WPIC, *Hydrogen Demand*; *Platinum and the Hydrogen Economy* (2023).



BlackRock iShares, *EEM Geographic Exposure* (Oct. 1, 2026).

Transmission Channel 3: South African platinum exports



South Africa is the world's dominant platinum producer, accounting for about 70% of global mined platinum supply

At the same time, South African companies make up roughly 3% of EEM

This creates a direct supply-side transmission channel. Stronger platinum prices can support South African mining revenues, exports, corporate earnings, and the rand, which can also support the South African portion of EEM

WPIC, *Platinum 101* (2020); BlackRock iShares, *EEM Geographic Exposure*; author's analysis.

Relationship & Economic Rationale

Why the link should persist

Hydrogen adoption could become a long-term structural trend, creating a more consistent source of platinum demand

AI investment may affect EEM more immediately, but over time continued AI infrastructure growth could also support platinum demand through electronics, semiconductors, sensors, and related hardware

South Africa should continues dominating global platinum production, especially if Russian supply remains constrained over the long term

The relationship could **weaken** if EEM becomes increasingly driven by AI and other non-platinum-related sectors, while platinum is dominated by separate commodity-specific supply and demand shocks

Key Numbers

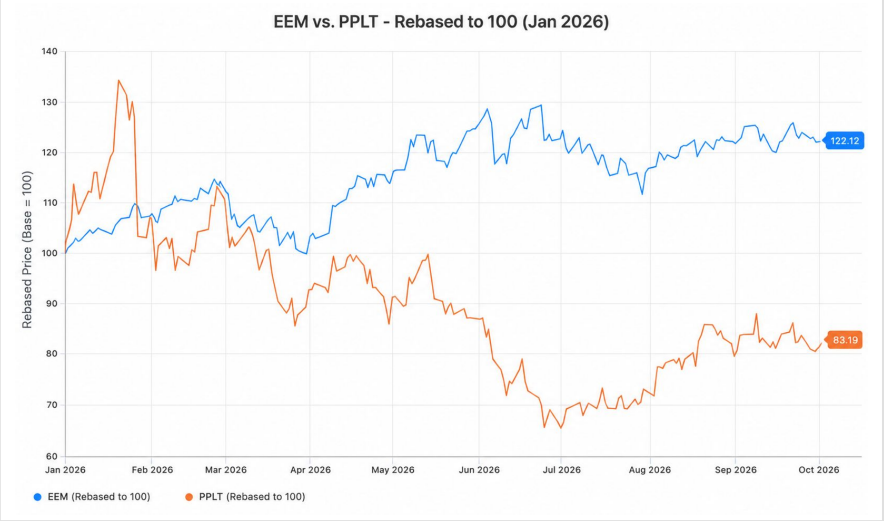
Asset Y	**EEM - Emerging Market Equities**
Asset X	**PPLT - Physical Platinum**
Data Range	**Jan. 8, 2010–Sept. 30, 2026**
Data Source	**Yahoo Finance**
Measure Used	**3-year rolling log-price regression residual**
Correlation (Full)	**+0.403**
Correlation (1Y)	**+0.451**
Historical average absolute gap	**8.48%**
Current Gap	**+18.63% above model's EEM level**
Current Z-Score	**+2.09σ**
Largest Past Gap (3 years)	**4.57σ**

Yahoo Finance, *EEM and PPLT Historical Prices*; WPIC, *Platinum Demand Drivers*; author's calculations and analysis.

Historical Evidence

How We Measured It

3-year rolling log regression of EEM on PPLT using daily adjusted prices (2010–2026). This measures how far EEM is trading from the level historically associated with platinum.



Yahoo Finance, *EEM and PPLT Historical Prices* (2010–2026); author's calculations.

Does the Relationship Hold Over Time?

Period	Correlation	Avg Gap (%)	Max Gap (σ)	Days ≥2σ	Corr ≥0.50?
Full Sample	0.403	8.49%	4.57σ	561	No
Last 10Y	0.425	9.40%	4.33σ	462	No
Last 5Y	0.412	8.09%	4.33σ	167	No
Last 3Y	0.419	8.25%	4.33σ	86	No
Last 1Y	0.452	10.60%	4.33σ	80	No
Dec 2025–Jan 2026 Shock	0.419	7.93%	2.78σ	5	No
Overall	0.403	8.49%	4.57σ	561	No

EEM and PPLT show a broadly consistent but moderate correlation of 0.40–0.45 across the reported windows, keeping a relatively consistent correlation during the PPLT spike

Normalization Rate

≥2σ Episodes	12-Month Cohort	Within 1M	Within 3M	Within 6M	Within 9M	Within 12M	Median Time*
9	8	0%	25%	62.5%	75%	75%	93 trading days

*Median includes only episodes that normalized; percentages use the common 8-episode cohort.

Normalized means return to < +0.5σ

Current Dislocation — Why Now?

WHY IS THE RELATIONSHIP DISLOCATED NOW?

Size of the Deviation

EEM is trading +2.09σ above the level implied by its recent relationship with platinum, equivalent to an ~18.6% relative premium.

What Caused It

The gap widened mainly because EEM benefited from a strong AI and semiconductor rally

Which Market Is Right?

AI infrastructure spending growth is expected to slow from nearly 100% in 2026 to about 37% in 2027 according to Reuters, which could ease EEM's AI-driven outperformance and support mean reversion in the EEM–PPLT relationship.

Asymmetric Setup

A move from the current +2.09σ gap to the +0.85σ target generates roughly +5.9% modeled upside, while widening to the +2.75σ stop implies about −3.4% downside, producing a 1.72x reward-to-risk ratio, while the trade targets 1.24σ of normalization versus only 0.66σ of further adverse widening,

NORMALIZATION CATALYSTS

0–1 Month — **Catalyst 1**

TSMC's Sept. sales on Oct. 8 and Q3 earnings on Oct. 15

[Expected: gap narrows to ~1.75σ]

2-5 Months — **Catalyst 2**

If Reuters prediction that AI spending growth is expected to slow from ~100% in 2026 to 37% in 2027 is confirmed, this could potentially cause a rotation out of semiconductor-heavy EEM and toward industrial assets.

[Expected: gap narrows to ~1.25σ]

6-12 Months — **Catalyst 3**

A broader shift from AI-led equity growth toward stronger global manufacturing and green hydrogen energy could allow the industrial cycle to reassert itself and narrow the EEM–PPLT gap.

[Expected: gap narrows to ~~0.85σ]

Reuters, *AI Spending Slowdown Concerns* (Sept. 15, 2026); TSMC, *Financial Calendar*; author's scenario analysis.

Market vs. Normal Relationship

What the Market Says vs. What the Relationship Says

Metric	Today	If Normal	Gap	Why It Should Close
EEM Level	66.79	56.30	+18.6%	EEM is trading well above its platinum-implied level, leaving room for mean reversion
Z-score	+2.09σ	0.0σ	+2.09σ	At +2.09σ, the relationship is unusually stretched relative to its recent history
Macro Story in X	Hydrogen Growth	—	—	Platinum demand should rise as the hydrogen economy continues to expand
Macro Story in Y	AI Growth	—	—	Slower AI investment growth could reduce EEM's tech-driven outperformance
Correlation	+0.45 (1y)	+0.40 (16 yr full history)	0.05	Recent correlation remains above its long-run level, suggesting the relationship is still intact.

Source of Edge

Why the Gap Exists

EEM is more heavily reliant on AI and semiconductor growth which has recently boomed, while PPLT is partially, but not nearly as heavily concentrated in this one industry.

Why It Hasn't Closed

AI investment is still strong and even increasing with new AI agents, while the hydrogen economy has not exploded quite yet.

What Proves You Right

AI investment cools, and/or broader industrial and hydrogen demand for platinum increases, causing the EEM-PPLT gap to narrow

Yahoo Finance, *Historical Prices*; WPIC, *Hydrogen Demand*; author's calculations and scenario analysis.

Trade Expression

POSITION PARAMETERS

Structure	**Long PPLT / Short EEM**
Instruments	**PLT ETF and EEM ETF**
Sizing	**37% long PPLT / 63% short EEM**
Entry	**+2.09σ (EEM $67.69 / PPLT $15.42)**
Target	**+0.85σ (+6.05%)**
Risk/Reward	**1.72x**
Stop	**2.75σ (–3.52%)**
Horizon	**1-6 months**
Exit Plan	**Take profit at +0.85σ; stop out at +2.75σ**

TRADE RATIONALE

Why This Expression

I'm not short EEM as a directional bearish view, rather I am shorting it as the relatively expensive leg of the pair to isolate the EEM–PPLT dislocation and reduce broader market exposure

Catalyst

Slower AI investment growth, stronger hydrogen demand, and stronger industrial demand could cause PPLT to outperform EEM in the short term

What Convergence Looks Like

PPLT outperforms EEM until the gap falls from +2.09σ toward +0.85σ

Cost of Holding

ETF fees, the cost of borrowing EEM to short it, and dividends owed on the short position

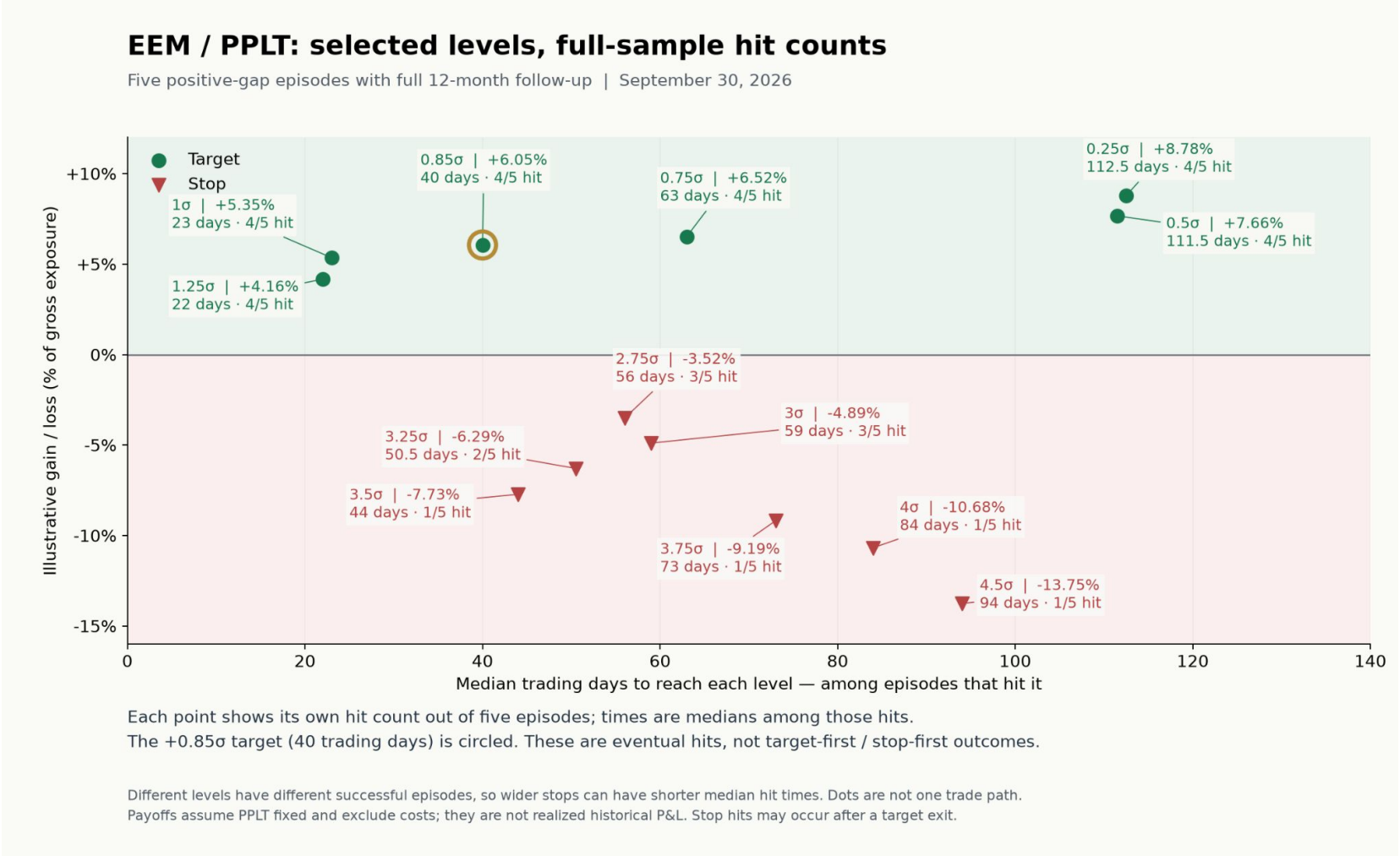
Liquidity

Both are exchange-traded ETFs, with EEM especially liquid and PPLT providing direct platinum exposure

Thesis Invalidation

The thesis fails if the gap reaches +2.75σ, the EEM/PPLT relationship materially breaks down, or AI-driven EEM strength proves a long term trend rather than a temporary one

Yahoo Finance; BlackRock iShares and Aberdeen Investments, *Fund Disclosures*; author's calculations.



Yahoo Finance, *EEM and PPLT Historical Prices*; author's historical episode and scenario analysis.

For a +0.85σ target

Stop	Target reached first	Stop reached first	Illustrative gain	Illustrative loss	R/R
+2.75σ	**4/5**	**1/5**	**6.05%**	**3.52%**	**1.72x**
+3.00σ	4/5	1/5	6.05%	4.89%	1.24x
+3.50σ	4/5	1/5	6.05%	7.73%	0.78x
+4.00σ	4/5	1/5	6.05%	10.68%	0.57x

Historical first-hit outcomes: five comparable positive-gap episodes, 12-month follow-up; daily closes.
Illustrative gain/loss: PPLT held fixed, percentage of gross exposure, before costs. Data as of September 30, 2026.

I avoided negative targets because too few historical episodes reached them to support reliable conclusions

Yahoo Finance, *EEM and PPLT Historical Prices*; author's historical episode and scenario analysis.

AI investment continues to increase

If the rise of AI agents continues to accelerate investment in semiconductors, data centers, and other AI infrastructure, platinum demand may also benefit, but likely not nearly as much as EEM because of its heavy technology exposure. In that case, EEM could keep outperforming and the gap could widen as the market enters a new structural regime.

Reuters, *AI Spending Slowdown Concerns* (Sept. 15, 2026); author's risk analysis.

Hydrogen demand disappoints

If PEM electrolyzers and FCEVs develop more slowly because government policy or weaker incentives reduce the shift toward clean energy, platinum demand may not rise as expected. That would make it harder for PPLT to catch up with EEM and close the gap.

WPIC, *Hydrogen Demand*; author's risk analysis.

Independent changes in platinum supply

If South African platinum production rises because of better mining technology or operating conditions, increased supply could limit platinum price gains. Likewise, if the Russia-Ukraine war ends and Russian platinum exports recover, global supply could increase while South Africa's relative importance declines, weakening one of the links between platinum and EEM.

WPIC, *Platinum 101* (2020); author's supply-risk scenarios.

Will AI continue to grow fast enough to outrun the broader industrial economy and justify what markets have already priced in?